



*IFRS17 vs SOLVENCY II
AND LOCAL GAAP APPROACHES TO
INSURANCE STATEMENTS:
A CASE STUDY*

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ABSTRACT

The “operational” translation of the principles of IFRS17 cannot ignore comparative analyses with the results of other regulatory principles. The Italian Institute for the Supervision of Insurance (IVASS) said that «[The] financial statement disclosure for statutory purposes, based on criteria defined by national legislation (Local Gaap), coexists with the market consistent valuations of assets and liabilities, required by the Solvency II regime, which entered into force in 2016, and the rules introduced by the IASB for the insurance market, where applicable». Therefore, it is necessary to regulate this “coexistence”.

In this paper, a logical (schematic, but essential) comparison structure is outlined, considering the effects on the profit for the year generated by insurance statements (financial statement, profit or loss) prepared by adopting the three “alternative (in the methods) but coexisting” approaches: Local Gaap, Solvency II and IFRS17.

It is a setting that is limited to a “statement arithmetic”, preliminary to detailed technical investigations yet relevant for the construction of valuation processes.

Keywords: IFRS17, Solvency II, Local Gaap, statement of financial position, statement of profit or loss

J.E.L. Classification: G10, G18, G22, G35, M48.

SUMMARY – Introduction. 1 Reference insurance portfolio, data used to compare insurance statements results. 2 Arithmetic of the insurance statements. 2.1 The reference insurance statements. 2.2 Definitions of quantities, relations among quantities. 3 Results. 4 Some considerations on the logic of the IFRS17 statements. Appendix – An example of insurance statement arithmetic (“case A”).

Introduction

Despite the extension of the documentation on the IFRS17 “project” (for insurance contracts) and the “illustrative examples”¹, there remains discretion on the interpretation of the principles, and there is a lively debate on the treatment of some contractual types (particularly relevant for the Italian market)².

On the other hand, the “operational” translation of the principles of IFRS17 cannot ignore comparative analyses with the results of other regulatory principles.

A premise of IVASS is relevant (and confirmatory): «[The] financial statement disclosure for statutory purposes, based on criteria defined by national legislation (Local Gaap), coexists with the market consistent valuations of assets and liabilities, required by the Solvency II regime, which entered into force in 2016, and the rules introduced by the IASB for the insurance market, where applicable»³.

It is necessary to regulate this “coexistence”, to bring the different evaluation methodologies to the maximum level of consistency, or to gain awareness in the interpretation of differences, in order to read the “true” status of the company (and – where appropriate – of the group). Otherwise, the “interpretative misalignment” (induced by different methodologies) makes the profitability assessment difficult, and the information base on which to develop tactical actions and strategies ambiguous.

¹ IFRS Foundation, *IFRS17 Insurance Contracts*. IFRS Standards, May 2017 [updated in June 2020]; IFRS Foundation, *IFRS17 Insurance Contracts. Illustrative Examples*, May 2017.

² It was noted that «L’IFRS 17 avrà notevoli impatti, dovuti principalmente alle regole da applicare ai contratti in essere al momento della transizione e al livello di aggregazione per la rilevazione dei contratti basato sul principio delle coorti annuali per cui, ai fini della misurazione delle passività e della rilevazione dei profitti, i contratti di ciascun portafoglio devono essere suddivisi in tre gruppi (onerosi, probabilmente non onerosi, profittevoli) all’interno dei quali non è possibile raggruppare contratti sottoscritti in annualità differenti. Il principio è stato oggetto di numerose osservazioni sulla scarsa coerenza con il modello di *business* assicurativo e sull’onerosità implementativa. L’EFRAG [l’European Financial Reporting Advisory] ha suggerito un’eccezione al principio delle coorti annuali per i contratti vita che prevedono una mutualità intergenerazionale e si è reso disponibile a fornire supporto allo IASB [l’International Accounting Standards Board] per una soluzione tecnica robusta e coerente con le caratteristiche del comparto» [IVASS, *Relazione sull’attività svolta dall’Istituto nell’anno 2019*, Roma, 18 giugno 2020, pagina 130]. The problem of protecting the principles of mutuality does not only concern the Italian market. Ania, together with the Associations of France, Spain, Denmark, the Czech Republic, Poland and Romania, sent a Joint Statement to EFRAG and the European Commission (on 9 July 2020): the European Commission and the Member States are requested «to develop a European solution» to the problem.

As for the timing, the entry into force of IFRS 17 is expected from 1 January 2023; from the same date, IFRS9 (relating to “Financial Instruments”) must be applied.

³ IVASS, *Relazione sull’attività svolta dall’Istituto nell’anno 2019*, Roma, 18 giugno 2020, pagina 41 («[l]’informativa di bilancio a fini civilistici, basata su criteri definiti dalla normativa nazionale (local gaap) coesiste con le valutazioni di attività e passività market consistent, richieste dal regime di solvibilità Solvency II, entrato in vigore nel 2016, e con le regole introdotte dallo IASB per il mercato assicurativo, ove applicabili»).

In this paper, a logical (schematic, but essential) comparison structure is outlined, considering the effects on the profit for the year generated by insurance statements (financial statement, profit or loss) prepared by adopting the three “alternative (in the methods) but coexisting” approaches: Local Gaap, Solvency II and IFRS17.

It is a setting that is limited to a “statement arithmetic”, preliminary to detailed technical investigations yet relevant for the construction of valuation processes (on how to deal with the contractual features of assets and liabilities, on how to formally represent and manage – in conditions of uncertainty – the market “variables”).

1 Reference insurance portfolio, data used to compare insurance statements results

The reference insurance portfolio – The reference insurance portfolio is a single premium portfolio, with a duration T of three years ($T = 3$), with benefits defined on an annual schedule, paid at the end of each year (at the end of the insurance statement years: $t = 1, 2, 3$). Premiums are all collected at the beginning of the time interval (in $t = 0$). In the IFRS17 approach, the initial valuation (*at inception*) includes the premiums due at time $t = 0$; and the reference insurance portfolio is the new business portfolio originates at the beginning of the first year.

The logic of the comparison – We set ourselves “at the end of the period”, *i.e.*, at the end of the third year. There is therefore “complete information” (*ex-ante* and *ex-post*) on different management situations. The description of the effects of the valuation logic becomes effective by comparing the outcome of different account profiles (being available – in order to be able to fulfill the interpretation of a fundamental criterion of IFRS17 – expected and actual cash flows).

Data – The data (cash flows) to be considered are represented by characterizing the amount with the due date (in some situations, to avoid ambiguity, it will be convenient to use the bivector form: “due amount / due time”). In some situations, it will also be necessary to indicate the valuation date (in which the expectation is elicited, and the calculation of the quantities is carried out).

For the purposes of the analysis, the reference insurance portfolio is described by three main types of cash flows:

1 – *actual cash inflow*, that in the case study it is a vector with one element (the premium payable in $t=0$):

$$\mathbf{X} = \{X_0^{act}\};$$

2 – *expected cash outflow*, *i.e.*, the set of amounts that are expected to be paid in the future.

There are two types of expected cash flow:

- a cash outflow valued at $t = 0$ with the technical assumptions used for initial pricing (I order assumption):

$$\bar{\mathbf{Y}}_0^I = \{\bar{Y}_{0,k}^I\}, \text{ amounts due at time } k = 1, 2, 3;$$

- a cash outflow valued with realistic technical assumptions (III order assumptions), at the beginning of each financial year for the remaining years. In the case considered there are therefore three vectors:

$$\bar{\mathbf{Y}}_t^{\text{III}} = \{\bar{Y}_{t,k}^{\text{III}}\}, \quad t = 0, 1, 2; \quad k = 1, 2, 3; \quad k > t;$$

3 – *actual cash outflow*, the set of amounts actually paid in each annual maturity:

$$\mathbf{Y} = \{Y_k^{\text{act}}\}, \quad k = 1, 2, 3.$$

It is assumed that the cash flows $\bar{\mathbf{Y}}_t^{\text{III}}$ (valued with realistic technical assumptions) include a risk loading for the insurance risk (which is equivalent to consider the risk margin in the Solvency II approach, and the risk adjustment in the IFRS17 approach).

Remark – Obviously, the reference portfolio also reduces the structure of the analysis scheme to essentiality. The types of cash flow considered do not exhaust the set of quantities to be evaluated, in a general scheme. For example, it is not considered in the IFRS17 approach and in the life sector, the distinction between investment and non-investment components of life benefits relating to insurance contracts with direct participation features and to death benefits⁴; it is not considered either the distinction, for annual premium insurance contracts without direct participation features, between current and future components of premiums, charges, commissions and other acquisition costs. □

Cash flow matrices – Tables 1, 2, 3, and 4 show the four cases of cash flow matrix analyzed.

Table 1 – cash flows “base case”

maturity	cash inflow	cash outflow				
	act	exp in 0 (l ord)	exp in 0 (real)	act/exp in 1 (real)	act/exp in 2 (real)	act in 3
0	30					
1		10	9	9		
2		10	8	8	8	
3		10	7	7	7	7

Table 2 – cash flows “case A”

maturity	cash inflow	cash outflow				
	act	exp in 0 (l ord)	exp in 0 (real)	act/exp in 1 (real)	act/exp in 2 (real)	act in 3
0	30					
1		10	9	8		
2		10	8	7	8	
3		10	7	6	7	7

⁴ IFRS Foundation, *IFRS17 Insurance Contracts*. IFRS Standards, May 2017 [updated in June 2020], paragraphs 10-13.

Table 3 – cash flows “case B”

maturity	cash inflow	cash outflow				
	act	exp in 0 (l ord)	exp in 0 (real)	act/exp in 1 (real)	act/exp in 2 (real)	act in 3
0	30					
1		10	9	10		
2		10	8	9	10	
3		10	7	8	9	9

Table 4 – cash flows “case C”

maturity	cash inflow	cash outflow				
	act	exp in 0 (l ord)	exp in 0 (real)	act/exp in 1 (real)	act/exp in 2 (real)	act in 3
0	30					
1		10	9	15		
2		10	8	15	10	
3		10	7	10	9	9

The “cash inflow” is the amount of the premium received by the insurer (in $t = 0$). The “cash outflow” columns show, for each date of the schedule, the amounts of the actual payments (“act”, in bold), and the expected payments (the columns mark the date on which the expectation is elicited: “in 0” – with first-order and realistic assumptions –, and then “in 1” and “in 2”).

The use of the bivector “due amounts/due time” is an unambiguous way to understand the formal representation of the cash flow. For example, with reference to “case A” (table 2) we have:

$$\mathbf{X} = \{30\}/\{0\}$$

$$\bar{\mathbf{Y}}_0^I = \{10, 10, 10\}/\{1, 2, 3\}$$

$$\bar{\mathbf{Y}}_0^{III} = \{9, 8, 7\}/\{1, 2, 3\}; \bar{\mathbf{Y}}_1^{III} = \{7, 6\}/\{2, 3\}; \bar{\mathbf{Y}}_2^{III} = \{7\}/\{3\}$$

$$\mathbf{Y} = \{8, 8, 7\}/\{1, 2, 3\}$$

On the differences in amounts – In the IFRS17’s jargon, the difference between expected and actual amounts due in the same date is defined as *current*; the difference between two expected amounts, previous minus subsequent, is defined as *future*.

Let us note that, in the four cases analyzed, the realistic cash outflows at $t = 0$ are equal. As regards the *current* and *future* differences in cash flows, we have that:

- in the “base case”, the differences are always equal to zero; in other words, the final actual/expected realistic cash flows (at the end of the period) are equal to the corresponding initial realistic expected cash flows (at the beginning of the period);
- in other cases, the final actual/expected realistic cash flows compared to the corresponding initial realistic expected cash flows are:
 - in “case A”, smaller in the first period (in $t = 1$ with respect to $t = 0$), greater in the second period (in $t = 2$ with respect to $t = 1$) and equal in the third period (in $t = 3$ with respect to $t = 2$);
 - in “case B”, greater in the first period (in $t=1$ compared to $t=0$), greater in the second period (in $t=2$ compared to $t=1$) and equal in the third period (in $t=3$ compared to $t=2$);

- in “case C” (the “extreme” case), significantly higher in the first period (in $t = 1$ with respect to $t = 0$), significantly lower in the second period (in $t = 2$ with respect to $t = 1$), equal in the third period (in $t = 3$ with respect to $t = 2$).

2 Arithmetic of the insurance statements

2.1 The reference insurance statements

Assumptions – It is assumed that: (a) the single premium is invested in liquidity, the insurance benefits are financed with liquidity (“Cash”), and any negative liquidity is fully financed; (b) the (annual) interest rate is equal to 0%; (c) the profit for the year is fully paid/financed (to/by shareholders); (d) expenses and costs are not considered. It is also assumed that, given the realistic expected cash flow in $t=0$ in the four cases analyzed, the reference portfolio is profitable at inception.

Remark. The simplifying assumption of a deterministic and zero interest rate “in any case” is justified only for the purpose of arguing on the logical (structural) relationships between quantities. It must remain alive in the background of the argument – without considering the role of the other risk drivers that affect the values – at least the multiplicity of meaning and role that the interest rate has in regulation and evaluation processes. Emblematic is the case, in life business, of profit-sharing policies with guaranteed minimum, where interest rates are concerned with: the “rate of return” of the asset portfolio (to which the policies are linked), the “guaranteed rate”, the “market rate” for the valuation of premiums and reserves; and where, in realistic valuations, the market rate and the rate of return have an intimate interconnection, and are managed by stochastic models⁵. □

Reference insurance statements – The reference insurance statement consists of the “profit or loss” statement and the “financial position” statement. Table 5 shows the stylized layouts of the insurance statements in the Local Gaap, IFRS17 and Solvency II approach (in the Solvency II approach only the profit or loss statement is significant)⁶.

⁵ De Felice, M., Moriconi, F., *Market Based Tools for Managing the Life Insurance Company*, Astin Bulletin, 35(2005), 1:

it is a reference that also illustrates in “operational” terms a model for the evolution of market risk drivers; the meaning to be given to the expressions “market consistent”, “valuation model”, “value of business in force” (VBIF), “measures of financial risk”; and the methods of valuation of insurance contracts with direct participation features.

⁶ The reference insurance statement aims to satisfy the purpose of the case study. In a more general setting for the IFRS17 approach, among the other significant quantities it could be considered, for example, the *Liability for Incurred Claims* (IFRS Foundation, *IFRS17 Insurance Contracts*. IFRS Standards, May 2017 [updated in June 2020], paragraphs 40-42).

Table 5 – reference insurance statements in the three approaches (Local Gaap, Solvency II, IFRS17)

Local Approach		
<i>Financial position</i>		<i>Profit or loss</i>
<u>Assets</u>	<u>Liabilities and Equity</u>	
Cash	Mathematical reserve	<u>Revenues</u>
		Premiums
		<u>Expenses</u>
		Actual claims
	Profit	Net reserve
		<u>Profit</u>

Solvency II Approach		
<i>Financial position</i>		<i>Profit or loss</i>
<u>Assets</u>	<u>Liabilities and Equity</u>	
Cash	Technical provisions	-
	Profit	

IFRS17 Approach		
<i>Financial position</i>		<i>Profit or loss</i>
<u>Assets</u>	<u>Liabilities and Equity</u>	
Cash	Fulfilment cash flows	<u>Revenues</u>
	Contractual Service Margin	Expected claims
	<u>Liability for Remaining Coverage</u>	CSM release
		<u>Expenses</u>
		Actual claims
	Profit	Loss component
		<u>Profit</u>

Remark. With respect to the three “accounting models” considered in the IFRS17 approach – Building Blocks Approach (BBA), Variable Fee Approach (VFA) and Premium Allocation Approach (PAA) – the reference insurance statements refers to the BBA model. □

2.2 Definitions of quantities, relations among quantities

On some reference documents – For definitions and comments of the relevant quantities, reference will be made to documents of IVASS (Regulation 22)⁷, Eiopa (Solvency II Directive)⁸, IFRS Foundation⁹.

⁷ ISVAP, *Regolamento n. 22*, 4th April 2008.

⁸ Directive 2009/138 / EC of the European Parliament and of the Council of 25th November 2009 (Solvency II), 12.01.2017.

⁹ IFRS Foundation, *IFRS17 Insurance Contracts*. IFRS Standards, May 2017 [updated in June 2020].

Relevant quantities – The relevant quantities are listed; for the comparison between “languages” and statement logics (as in use, the subscript is used for the reference time):

R_t “Mathematical reserve”;

principles and some methods of valuation of reserves (mathematical, technical) for the insurance statements under the Local Gaap approach (also called Solvency I approach) are prescribed in Regulation 22 of the IVASS.

TP_t “Technical provisions”;

The “Technical Provisions” (TP) are defined in Article 76 of the Solvency II Directive: their value must be equal to the amount needed by the company to transfer the existing obligations to another company. They are given by the sum of the Best estimate and Risk margin [article 77]. The “Best Estimate” (BE) must be calculated as «probability-weighted average of future cash-flows, taking account of the time value of money (expected present value of future cash-flows), using the relevant risk-free interest rate term structure, [...] up-to-date and credible information and realistic assumptions and be performed using adequate, applicable and relevant actuarial and statistical methods» [article 77]. The “Risk Margin” (RM) must be such as to ensure that «the value of the technical provisions is equivalent to the amount that insurance [...] undertakings would be expected to require in order to take over and meet [...] obligations».

Remark. Matching between logical categories: the “Technical Provisions” match the “reserves” of Solvency I. □

Remark. The Solvency II “Risk Margin” mates the IFRS17 “risk adjustment for non-financial risk”. In the IFRS17 approach the “risk adjustment for non-financial risk” regards «[the] compensation an entity requires for bearing the uncertainty about the amount and timing of the cash flows that arises from non-financial risk as the entity fulfills insurance contracts.». □

FCF_t “Fulfilment cash flows”;

«An explicit, unbiased and probability-weighted estimate (i.e. expected value) of the present value of the future cash outflows minus the present value of the future cash inflows that will arise as the entity fulfills insurance contracts, including a risk adjustment for non-financial risk.».

CSM_t “Contractual Service Margin”;

«A component of the carrying amount of the asset or liability for a group of insurance contracts representing the unearned profit the entity will recognize as it provides insurance contract services under the insurance contracts in the group.».

LRC_t “Liability for Remaining Coverage”;

«An entity’s obligation to: (a) investigate and pay valid claims under existing insurance contracts for insured events that have not yet occurred (i.e. the obligation that relates to the unexpired portion of the insurance coverage); and (b) pay amounts under existing insurance contracts that are not included in (a) and that relate to: (i) insurance contract services not yet provided (i.e. the obligations that relate to future provision of insurance contract services); or (ii) any investment components or other amounts that are not related to the provision of insurance contract services and that have not been transferred to the liability for incurred claims.».

C_t “Cash”;

it is the liquidity balance in t .

ΔR_t “Net reserve”;

it is the difference between the mathematical reserve in $t + 1$ and in t .

ΔCSM_t^* “CSM release”;

it is the share of CSM charged to the profit or loss statement for year t .

L_t “Loss component”;

if the CSM value before “CSM release” is negative, it is the value charged to the profit or loss statement for year t .

U_t “Profit”;

it is the profit for the financial year t ; if negative it is a loss.

Formalization of the quantities - Given the assumptions and the reference insurance statements we have:

- in the Local Gaap approach:

$$R_t = \sum_{k>t} Y_{0,k}^I,$$

$$\Delta R_t = R_t - R_{t-1}$$

$$C_1 = X_0^{act} - Y_1^{act}, \quad C_{t:t>1} = C_{t-1} - Y_t^{act} - U_{t-1}$$

$$U_1 = X_0^{act} - Y_1^{act} - \Delta R_1 = C_1 - R_1, \quad U_{t:t>1} = -Y_t^{act} - \Delta R_t = C_t - R_t$$

- in the Solvency II approach:

$$TP_t = \sum_{k>t} \bar{Y}_{t,k}^{III}$$

$$C_1 = X_0^{act} - Y_1^{act}, \quad C_{t:t>1} = C_{t-1} - Y_t^{act} - U_{t-1}$$

$$U_t = C_t - TP_t$$

- in the IFRS17 approach:

$$FCF_t = \sum_{k>t} \bar{Y}_{t,k}^{III}, \quad FCF_0 = \sum_{k>0} \bar{Y}_{0,k}^{III} - X_0^{act}$$

$$\Delta FCF_t^{future} = \sum_{k>t} \bar{Y}_{t-1,k}^{III} - \bar{Y}_{t,k}^{III}$$

$$\Delta FCF_t^{current} = \bar{Y}_{t-1,t}^{III} - Y_t^{act}$$

$$CSM_t^* = \max \{ CSM_{t-1} + \Delta FCF_t^{future}; 0 \}, \quad CSM_0 = \max \{ -FCF_0; 0 \}$$

$$\Delta CSM_t^* = CSM_t^* / (T - t + 1)$$

$$CSM_t = CSM_t^* - \Delta CSM_t^*$$

$$L_t = -\min \{ CSM_{t-1} + \Delta FCF_t^{future}; 0 \}, \quad L_0 = -\min \{ -FCF_0; 0 \}$$

$$LRC_t = FCF_t + CSM_t, \quad LRC_0 = FCF_0 + CSM_0$$

$$C_1 = X_0^{act} - Y_1^{act}, \quad C_{t:t>1} = C_{t-1} - Y_t^{act} - U_{t-1}$$

$$U_t = \bar{Y}_{t-1,t}^{III} + \Delta CSM_t^* - Y_t^{act} - L_t = C_t - LRC_t$$

Quantities and cash flow - Quantities depend on:

- in the Local Gaap approach:
 - *actual cash inflow/outflow*,
 - differences between *I order expected cash outflow* at the end and at the beginning of each financial year;
- in the Solvency II approach:
 - *actual cash inflow/outflow*,
 - *expected realistic cash outflow* at the end of each financial year;
- in the IFRS17 approach:
 - *actual cash inflow/outflow*,
 - *current difference*,
 - *future difference*.

Components of the Fulfilment Cash Flow – In the IFRS17 approach, for example considering the cash flows of “case A” (Table 2), the current and future differences of the FCF in $t = 1$ are:

$$\begin{aligned}\Delta FCF_1^{current} &= \bar{Y}_{0,1}^{III} - Y_1^{act} = 9 - 8 = 1 \\ \Delta FCF_1^{future} &= (\bar{Y}_{0,2}^{III} + \bar{Y}_{0,3}^{III}) - (\bar{Y}_{1,2}^{III} + \bar{Y}_{1,3}^{III}) = (8 + 7) - (7 + 6) = 2\end{aligned}$$

On profitability assessment – In the IFRS17 approach, in the four cases analyzed and at the beginning of the period ($t=0$), we have:

$$CSM_0 = \max \{-FCF_0; 0\} = \max \{-(7 + 8 + 9 - 30); 0\} = 6.$$

In the jargon of the principle, CSM_0 is the *CSM at inception*. The valuation of *CSM at inception*, realized in the so called “onerous contracts recognition test”, qualifies an insurance contract as “profitable” (if positive) or “onerous” (if negative). In the case study, the reference insurance portfolio is profitable.

In the framework of profitability assessment, a logical match is between the *CSM at inception* and the *New Business Value* at the same time. Note that, unlike the typical profitability assessment carried out in each financial year, in the IFRS17 approach the profitability of the insurance contract is assessed only at the initial date (in $t=0$) and, subsequently, on a recurring basis. For example in the “case A” and in $t = 1$, we have:

$$CSM_1^* = 8 \quad \Delta CSM_t^* = 2,7 \quad CSM_1 = 8 - 2,7 = 5,3 \quad VBIF_1 = 9$$

Level of aggregation of insurance contracts – In the IFRS17 approach, for the purposes of aggregating insurance contracts the reference portfolio is qualified as profitable and belonging to cohort 1¹⁰.

¹⁰ IFRS Foundation, *IFRS17 Insurance Contracts*. IFRS Standards, May 2017 [updated in June 2020] (paragraphs 14-24).

3 Results

Profit for the year, arithmetic results – Tables 6, 7, 8, and 9 show and give a summary view of the trend in profit for the year corresponding, respectively, to the four cases of expected and actual cash flows (“base case”, “case A”, “case B”, “case C”), in the three statement approaches analyzed (“Local”, “IFRS17”, “Solvency II”). Tables 10, 11, 12, and 13 show the results derived by the statements arithmetic (in the four cases of cash flow and in the three approaches). Results are presented with one decimal place.

Table 6 – “base case” profits’ trend

<i>financial year</i>	<i>Local</i>	<i>IFRS17</i>	<i>Solvency II</i>
1	1,0	2,0	6,0
2	2,0	2,0	0,0
3	3,0	2,0	0,0
Total profit	6,0	6,0	6,0

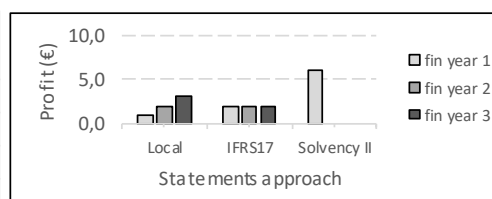


Table 7 – “case A” profits’ trend

<i>financial year</i>	<i>Local</i>	<i>IFRS17</i>	<i>Solvency II</i>
1	2,0	3,7	9,0
2	2,0	1,2	-2,0
3	3,0	2,2	0,0
Total profit	7,0	7,0	7,0

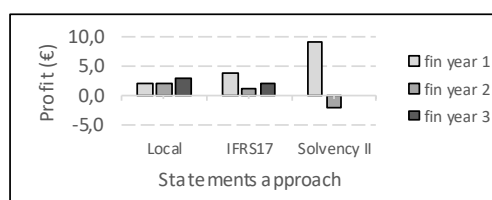


Table 8 – “case B” profits’ trend

<i>financial year</i>	<i>Local</i>	<i>IFRS17</i>	<i>Solvency II</i>
1	0,0	0,3	3,0
2	0,0	-0,2	-2,0
3	1,0	0,8	0,0
Total profit	1,0	1,0	1,0

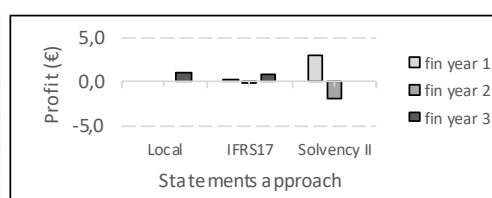
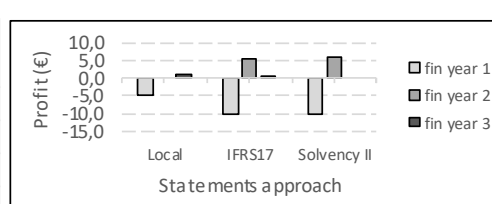


Table 9 – “case C” profits’ trend

<i>financial year</i>	<i>Local</i>	<i>IFRS17</i>	<i>Solvency II</i>
1	-5,0	-10,0	-10,0
2	0,0	5,5	6,0
3	1,0	0,5	0,0
Total profit	-4,0	-4,0	-4,0



An example of insurance statement arithmetic (in “case A”) is given in the appendix.

On the distribution of total profit (ex-post) – Note that:

- “Total profit” (defined as the ex-post difference between the actual premium received and the three benefits actually paid) is independent of the statement approach: in the first three cases (“base case”, “case A”, “case B”) is positive and equal to 6, 7, 1 respectively; in “case C”, however, it is negative (it is a total loss);
- in each of the four cases, the distribution of the total profit over the annual schedule varies as the statement approach varies: in the first three cases (with

total positive profit) the distribution in the IFRS17 approach is more *in line* with that of the Local, the more the expected and actual flows are similar; in “case C” (total loss) the distribution in the IFRS17 approach is, however, more *in line* with that of the Solvency II approach;

- in the “base case” (of *perfect forecast*, in which the cash flow *current* and *future* differences are always equal to zero), *maximum stability* in the profit trend is achieved; the more the differences are accentuated, the more the stability of the trend is weakened.

Table 10 – results produced with insurance statements arithmetic in the “base case”

Input: reference portfolio cash flow (ex-post)						
maturity	cash inflow	cash outflow				
	act	exp in 0 (l ord)	exp in 0 (real)	act/exp in 1 (real)	act/exp in 2 (real)	act in 3
0	30					
1		10	9	9		
2		10	8	8	8	
3		10	7	7	7	7

Output: statement of profit or loss

	financial year 1		financial year 2		financial year 3	
	Local	IFRS17	Local	IFRS17	Local	IFRS17
Revenues (Local/IFRS17)	30,0	11,0	0,0	10,0	0,0	9,0
Premiums	30,0		0,0		0,0	
Expected claims		9,0		8,0		7,0
CSM release		2,0		2,0		2,0
Expenses (Local/IFRS17)	29,0	9,0	-2,0	8,0	-3,0	7,0
Actual claims	9,0		8,0		7,0	
Net reserve	20,0		-10,0		-10,0	
Actual claims		9,0		8,0		7,0
Loss component		0,0		0,0		0,0
Profit	1,0	2,0	2,0	2,0	3,0	2,0

Output: trend for Total profit

financial year	Local	IFRS17	Solvency II
1	1,0	2,0	6,0
2	2,0	2,0	0,0
3	3,0	2,0	0,0
Total profit	6,0	6,0	6,0

Profit (€)

LocalIFRS17Solvency II

Statements approach

fin year 1

fin year 2

fin year 3

Output: statement of financial position

Assets				Liabilities and Equity			
t=1							
	Local	Solvency II	IFRS17		Local	Solvency II	IFRS17
C ₁	21,0	21,0	21,0	R ₁	20,0		
				TP ₁		15,0	
				FCF ₁			15,0
				CSM ₁			4,0
				LRC ₁			19,0
				U ₁	1,0	6,0	2,0
Total	21,0	21,0	21,0	Total	21,0	21,0	21,0
t=2							
	Local	Solvency II	IFRS17		Local	Solvency II	IFRS17
C ₂	12,0	7,0	11,0	R ₂	10,0		
				TP ₂		7,0	
				FCF ₂			7,0
				CSM ₂			2,0
				LRC ₂			9,0
				U ₂	2,0	0,0	2,0
Total	12,0	7,0	11,0	Total	12,0	7,0	11,0
t=3							
	Local	Solvency II	IFRS17		Local	Solvency II	IFRS17
C ₃	3,0	0,0	2,0	R ₃	0,0		
				TP ₃		0,0	
				FCF ₃			0,0
				CSM ₃			0,0
				LRC ₃			0,0
				U ₃	3,0	0,0	2,0
Total	3,0	0,0	2,0	Total	3,0	0,0	2,0

Output: IFRS17 assessment

t=0	
FCF ₀	-6,0
CSM ₀	6,0
LRC ₀	0,0
t=1	
CSM ₀	6,0
ΔFCF _{1 future}	0,0
CSM ₁ *	6,0
ΔCSM ₁ *	2,0
CSM ₁	4,0
L ₁	0,0
ΔFCF _{1 current}	0,0
t=2	
CSM ₁	4,0
ΔFCF _{2 future}	0,0
CSM ₂ *	4,0
ΔCSM ₂ *	2,0
CSM ₂	2,0
L ₂	0,0
ΔFCF _{2 current}	0,0
t=3	
CSM ₂	2,0
ΔFCF _{3 future}	0,0
CSM ₃ *	2,0
ΔCSM ₃ *	2,0
CSM ₃	0,0
L ₃	0,0
ΔFCF _{3 current}	0,0

Table 11 – results produced with insurance statements arithmetic in the “case A”

Input: reference portfolio cash flow (ex-post)						
maturity	cash inflow		cash outflow			
	act	exp in 0 (l ord)	exp in 0 (real)	act/exp in 1 (real)	act/exp in 2 (real)	act in 3
0	30					
1		10	9	8		
2		10	8	7	8	
3		10	7	6	7	7

Output: statement of profit or loss

	financial year 1		financial year 2		financial year 3	
	Local	IFRS17	Local	IFRS17	Local	IFRS17
Revenues (Local//IFRS17)	30,0	11,7	0,0	9,2	0,0	9,2
Premiums	30,0		0,0		0,0	
Expected claims		9,0		7,0		7,0
CSM release		2,7		2,2		2,2
Expenses (Local//IFRS17)	28,0	8,0	-2,0	8,0	-3,0	7,0
Actual claims	8,0		8,0		7,0	
Net reserve	20,0		-10,0		-10,0	
Actual claims		8,0		8,0		7,0
Loss component		0,0		0,0		0,0
Profit	2,0	3,7	2,0	1,2	3,0	2,2

Output: trend for Total profit

financial year	Local	IFRS17	Solvency II
1	2,0	3,7	9,0
2	2,0	1,2	-2,0
3	3,0	2,2	0,0
Total profit	7,0	7,0	7,0

Profit (€)

Local

IFRS17

Solvency II

Statements approach

Output: statement of financial position

Assets				Liabilities and Equity			
t=1							
	Local	Solvency II	IFRS17		Local	Solvency II	IFRS17
C ₁	22,0	22,0	22,0	R ₁	20,0		
				TP ₁		13,0	
				FCF ₁			13,0
				CSM ₁			5,3
				LRC ₁			18,3
				U ₁	2,0	9,0	3,7
Total	22,0	22,0	22,0	Total	22,0	22,0	22,0
t=2							
	Local	Solvency II	IFRS17		Local	Solvency II	IFRS17
C ₂	12,0	5,0	10,3	R ₂	10,0		
				TP ₂		7,0	
				FCF ₂			7,0
				CSM ₂			2,2
				LRC ₂			9,2
				U ₂	2,0	-2,0	1,2
Total	12,0	5,0	10,3	Total	12,0	5,0	10,3
t=3							
	Local	Solvency II	IFRS17		Local	Solvency II	IFRS17
C ₃	3,0	0,0	2,2	R ₃	0,0		
				TP ₃		0,0	
				FCF ₃			0,0
				CSM ₃			0,0
				LRC ₃			0,0
				U ₃	3,0	0,0	2,2
Total	3,0	0,0	2,2	Total	3,0	0,0	2,2

Output: IFRS17 assessment

t=0	
FCF ₀	-6,0
CSM ₀	6,0
LRC ₀	0,0
t=1	
CSM ₀	6,0
ΔFCF ₁ future	2,0
CSM ₁ *	8,0
ΔCSM ₁ *	2,7
CSM ₁	5,3
L ₁	0,0
ΔFCF ₁ current	1,0
t=2	
CSM ₁	5,3
ΔFCF ₂ future	-1,0
CSM ₂ *	4,3
ΔCSM ₂ *	2,2
CSM ₂	2,2
L ₂	0,0
ΔFCF ₂ current	-1,0
t=3	
CSM ₂	2,2
ΔFCF ₃ future	0,0
CSM ₃ *	2,2
ΔCSM ₃ *	2,2
CSM ₃	0,0
L ₃	0,0
ΔFCF ₃ current	0,0

Table 12 – results produced with insurance statements arithmetic in the “case B”

Input: reference portfolio cash flow (ex-post)						
maturity	cash inflow	cash outflow				
	act	exp in 0 (l ord)	exp in 0 (real)	act/exp in 1 (real)	act/exp in 2 (real)	act in 3
0	30					
1		10	9	10		
2		10	8	9	10	
3		10	7	8	9	9

Output: statement of profit or loss

	financial year 1		financial year 2		financial year 3	
	Local	IFRS17	Local	IFRS17	Local	IFRS17
Revenues (Local/IFRS17)	30,0	10,3	0,0	9,8	0,0	9,8
Premiums	30,0		0,0		0,0	
Expected claims		9,0		9,0		9,0
CSM release		1,3		0,8		0,8
Expenses (Local/IFRS17)	30,0	10,0	0,0	10,0	-1,0	9,0
Actual claims	10,0		10,0		9,0	
Net reserve	20,0		-10,0		-10,0	
Actual claims		10,0		10,0		9,0
Loss component		0,0		0,0		0,0
Profit	0,0	0,3	0,0	-0,2	1,0	0,8

Output: trend for Total profit

financial year	Local	IFRS17	Solvency II
1	0,0	0,3	3,0
2	0,0	-0,2	-2,0
3	1,0	0,8	0,0
Total profit	1,0	1,0	1,0

Profit (€)

Local

IFRS17

Solvency II

Statements approach

Output: statement of financial position

Assets				Liabilities and Equity			
t=1							
	Local	Solvency II	IFRS17		Local	Solvency II	IFRS17
C ₁	20,0	20,0	20,0	R ₁	20,0		
				TP ₁		17,0	
				FCF ₁			17,0
				CSM ₁			2,7
				LRC ₁		19,7	
				U ₁	0,0	3,0	0,3
Total	20,0	20,0	20,0	Total	20,0	20,0	20,0
t=2							
	Local	Solvency II	IFRS17		Local	Solvency II	IFRS17
C ₂	10,0	7,0	9,7	R ₂	10,0		
				TP ₂		9,0	
				FCF ₂			9,0
				CSM ₂			0,8
				LRC ₂		9,8	
				U ₂	0,0	-2,0	-0,2
Total	10,0	7,0	9,7	Total	10,0	7,0	9,7
t=3							
	Local	Solvency II	IFRS17		Local	Solvency II	IFRS17
C ₃	1,0	0,0	0,8	R ₃	0,0		
				TP ₃		0,0	
				FCF ₃			0,0
				CSM ₃			0,0
				LRC ₃		0,0	
				U ₃	1,0	0,0	0,8
Total	1,0	0,0	0,8	Total	1,0	0,0	0,8

Output: IFRS17 assessment

t=0	
FCF ₀	-6,0
CSM ₀	6,0
LRC ₀	0,0
t=1	
CSM ₀	6,0
ΔFCF ₁ future	-2,0
CSM ₁ *	4,0
ΔCSM ₁ *	1,3
CSM ₁	2,7
L ₁	0,0
ΔFCF ₁ current	-1,0
t=2	
CSM ₁	2,7
ΔFCF ₂ future	-1,0
CSM ₂ *	1,7
ΔCSM ₂ *	0,8
CSM ₂	0,8
L ₂	0,0
ΔFCF ₂ current	-1,0
t=3	
CSM ₂	0,8
ΔFCF ₃ future	0,0
CSM ₃ *	0,8
ΔCSM ₃ *	0,8
CSM ₃	0,0
L ₃	0,0
ΔFCF ₃ current	0,0

Table 13 – results produced with insurance statements arithmetic in the “case C”

Input: reference portfolio cash flow (ex-post)						
maturity	cash inflow	cash outflow				
	act	exp in 0 (l ord)	exp in 0 (real)	act/exp in 1 (real)	act/exp in 2 (real)	act in 3
0	30					
1		10	9	15		
2		10	8	15	10	
3		10	7	10	9	9

Output: statement of profit or loss						
	financial year 1		financial year 2		financial year 3	
	Local	IFRS17	Local	IFRS17	Local	IFRS17
Revenues (Local/IFRS17)	30,0	9,0	0,0	15,5	0,0	9,5
Premiums	30,0		0,0		0,0	
Expected claims		9,0		15,0		9,0
CSM release		0,0		0,5		0,5
Expenses (Local/IFRS17)	35,0	19,0	0,0	10,0	-1,0	9,0
Actual claims	15,0		10,0		9,0	
Net reserve	20,0		-10,0		-10,0	
Actual claims		15,0		10,0		9,0
Loss component		4,0		0,0		0,0
Profit	-5,0	-10,0	0,0	5,5	1,0	0,5

Output: trend for Total profit			
financial year	Local	IFRS17	Solvency II
1	-5,0	-10,0	-10,0
2	0,0	5,5	6,0
3	1,0	0,5	0,0
Total profit	-4,0	-4,0	-4,0

Profit (€)

Local IFRS17 Solvency II

Statements approach

■ fin year 1
■ fin year 2
■ fin year 3

Output: statement of financial position								
Assets				Liabilities and Equity				
t=1								
	Local	Solvency II	IFRS17		Local	Solvency II	IFRS17	
C ₁	15,0	15,0	15,0	R ₁	20,0			
				TP ₁		25,0		
				FCF ₁			25,0	
				CSM ₁			0,0	
				LRC ₁			25,0	
				U ₁	-5,0	-10,0	-10,0	
Total	15,0	15,0	15,0	Total	15,0	15,0	15,0	
t=2								
	Local	Solvency II	IFRS17		Local	Solvency II	IFRS17	
C ₂	10,0	15,0	15,0	R ₂	10,0			
				TP ₂		9,0		
				FCF ₂			9,0	
				CSM ₂			0,0	
				LRC ₂			9,0	
				U ₂	0,0	6,0	5,0	
Total	10,0	15,0	15,0	Total	10,0	15,0	15,0	
t=3								
	Local	Solvency II	IFRS17		Local	Solvency II	IFRS17	
C ₃	1,0	0,0	0,5	R ₃	0,0			
				TP ₃		0,0		
				FCF ₃			0,0	
				CSM ₃			0,0	
				LRC ₃			0,0	
				U ₃	1,0	0,0	0,5	
Total	1,0	0,0	0,5	Total	1,0	0,0	0,5	

Output: IFRS17 assessment	
t=0	
FCF ₀	-6,0
CSM ₀	6,0
LRC ₀	0,0
t=1	
CSM ₀	6,0
ΔFCF ₁ future	-10,0
CSM ₁ *	0,0
ΔCSM ₁ *	0,0
CSM ₁	0,0
L ₁	4,0
ΔFCF ₁ current	-6,0
t=2	
CSM ₁	0,0
ΔFCF ₂ future	1,0
CSM ₂ *	1,0
ΔCSM ₂ *	0,5
CSM ₂	0,5
L ₂	0,0
ΔFCF ₂ current	5,0
t=3	
CSM ₂	0,5
ΔFCF ₃ future	0,0
CSM ₃ *	0,5
ΔCSM ₃ *	0,0
CSM ₃	0,0
L ₃	0,0
ΔFCF ₃ current	0,0

4 Some considerations on the logic of the IFRS17 statements

The role of the forecast - In the logical structure of IFRS17, *forecast* has a significant role and effects.

The results of the insurance statements are affected by the differences between expected values (relating to the same due date, elicited on different dates, therefore with different information base); and between expected and actual values (relating to the same due date).

In detail, the amount of revenues and costs, and the final value of balance sheet items (such as the CSM's components and – obviously – the CSM) depend, respectively, on the comparison between:

- initial forecast (*ex-ante*) and realization (*ex-post*) of the cash flow for the current year (*current* difference);
- initial forecast and final forecast of the cash flow relating to the remaining years (*future* difference).

Remark – The comparison between *expected value* (outcome of the forecast) and *actual or realized value* should be considered with great attention. It plays a role in the foundations of probability theory: it enters the formal structure of the “criteria for evaluations”, while respecting consistency; and of the “decision theory” since it invokes the criticism of judgment (on the outcome of the prediction) vitiated by *hindsight*.

Not negligible – due to the potential effects on corporate governance – is the effect classified (in the jargon of behavioralism) as a “result bias”: so «[the] decision makers who foresee that their decisions will be analyzed with hindsight are induced to find bureaucratic solutions»¹¹.

In the case of the management of the insurance company, the problem is exacerbated by the “mixture” that occurs in the elicitation of the expectation among *technical bases of experience*, the role of the *expert opinion*, and *pre-defined algorithmic structure* (as it is in the Solvency II regulations). □

On the “management” of the balance sheet – With the principles of IFRS17, the balance sheet result (ultimately the profit) depends on the ability to *forecast*: in fact, it depends on *current* and *future* differences.

Aware of de Finetti's wise teaching on «prediction is not prediction», if we want stability we will need – even in the case of coherent predictions, formulated *at best*, with the *best effort* –: to protect ourselves from not having “guessed” the future (in a certain sense, to face “bad luck”).

Forms of reinsurance, properly designed, could be the means to reach this end.

¹¹ de Finetti, B., *Teoria delle probabilità*, Torino, Einaudi, 1970: on consistency pages 89-90, 101-114; on «*hindsight*» pages 246, 244. As far as «result bias» is concerned, see Daniel Kahneman, in *Pensieri lenti e veloci*, Milano, Mondadori, 2012, pages 224-225.

Appendix – An example of insurance statement arithmetic (“case A”)

- Local Gaap approach:

$$\underline{t = 1}$$

$$R_1 = Y_{0,1}^I + Y_{0,2}^I = 10 + 10 = 20$$

$$\Delta R_1 = R_1 - R_0 = 20 - 0 = 20$$

$$C_1 = X_0^{act} - Y_1^{act} = 30 - 8 = 22$$

$$U_1 = X_0^{act} - Y_1^{act} - \Delta R_1 = 30 - 8 - 20 = C_1 - R_1 = 22 - 20 = 2$$

$$\underline{t = 2}$$

$$R_2 = Y_2^I = 10 = 10$$

$$\Delta R_2 = R_2 - R_1 = 10 - 20 = -10$$

$$C_2 = C_1 - Y_2^{act} - U_1 = 22 - 8 - 2 = 12$$

$$U_2 = -Y_2^{act} - \Delta R_2 = -8 - (-10) = C_2 - R_2 = 12 - 10 = 2$$

$$\underline{t = 3}$$

$$R_3 = 0$$

$$\Delta R_3 = R_3 - R_2 = 0 - 10 = -10$$

$$C_3 = C_2 - Y_3^{act} - U_2 = 12 - 7 - 2 = 3$$

$$U_3 = -Y_3^{act} - \Delta R_3 = -7 - (-10) = C_3 - R_3 = 3 - 0 = 3$$

- Solvency II approach:

$$\underline{t = 1}$$

$$TP_1 = \bar{Y}_{1,2}^{III} + \bar{Y}_{1,3}^{III} = 7 + 6 = 13$$

$$C_1 = X_0^{act} - Y_1^{act} = 30 - 8 = 22$$

$$U_1 = C_1 - TP_1 = 22 - 13 = 9$$

$$\underline{t=2}$$

$$TP_2 = \bar{Y}_{2,3}^{III} = 7$$

$$C_2 = C_1 - Y_2^{act} - U_1 = 22 - 8 - 9 = 5$$

$$U_2 = C_2 - TP_2 = 5 - 7 = -2$$

$$\underline{t=3}$$

$$TP_3 = 0$$

$$C_3 = C_2 - Y_3^{act} - U_2 = 5 - 7 - (-2) = 0$$

$$U_3 = C_3 - TP_3 = 0 - 0 = 0$$

- IFRS17 approach:

$$\underline{t = 0}$$

$$FCF_0 = \bar{Y}_{0,1}^{III} + \bar{Y}_{0,2}^{III} + \bar{Y}_{0,3}^{III} - X_0^{act} = 9 + 8 + 7 - 30 = -6$$

$$CSM_0 = \max \{-FCF_0; 0\} = \max \{-(-6); 0\} = 6$$

$$L_0 = -\min \{-FCF_0; 0\} = -\min \{-(-6); 0\} = 0$$

$$LRC_0 = FCF_0 + CSM_0 = -6 + 6 = 0$$

$$\underline{t = 1}$$

$$FCF_1 = \bar{Y}_{1,2}^{III} + \bar{Y}_{1,3}^{III} = 7 + 6 = 13$$

$$\Delta FCF_1^{future} = \sum_{k>1} \bar{Y}_{0,k}^{III} - \bar{Y}_{1,k}^{III} = (8 + 7) - (7 + 6) = 2$$

$$\Delta FCF_1^{current} = \bar{Y}_{0,1}^{III} - Y_1^{act} = 9 - 8 = 1$$

$$CSM_1^* = \max \{ CSM_0 + \Delta FCF_1^{future}; 0 \} = \max \{ 6 + 2; 0 \} = 8$$

$$\Delta CSM_1^* = CSM_1^* / (T - t + 1) = 8 / (3 - 1 + 1) = 2,7$$

$$CSM_1 = CSM_1^* - \Delta CSM_1^* = 8 - 2,7 = 5,3$$

$$L_1 = -\min \{ CSM_0 + \Delta FCF_1^{future}; 0 \} = -\min \{ 6 + 2; 0 \} = 0$$

$$LRC_1 = FCF_1 + CSM_1 = 13 + 5,3 = 18,3$$

$$C_1 = X_0^{act} - Y_1^{act} = 30 - 8 = 22$$

$$U_1 = \bar{Y}_{0,1}^{III} + \Delta CSM_1^* - Y_1^{act} - L_1 = 9 + 2,7 - 8 - 0 = C_1 - LRC_1 = 22 - 18,3 = 3,7$$

$$\underline{t = 2}$$

$$FCF_2 = \bar{Y}_{2,3}^{III} = 7 = 7$$

$$\Delta FCF_2^{future} = FCF_1^{future} - FCF_2^{future} = 6 - 7 = -1$$

$$\Delta FCF_2^{current} = \bar{Y}_{1,2}^{III} - Y_2^{act} = 7 - 8 = -1$$

$$CSM_2^* = \max \{ CSM_1 + \Delta FCF_2^{future}; 0 \} = \max \{ 5,3 - 1; 0 \} = 4,3$$

$$\Delta CSM_2^* = CSM_2^* / (T - t + 1) = 4,3 / (3 - 2 + 1) = 2,2$$

$$CSM_2 = CSM_2^* - \Delta CSM_2^* = 4,3 - 2,2 = 2,2$$

$$L_2 = -\min \{ CSM_1 + \Delta FCF_2^{future}; 0 \} = -\min \{ 5,3 - 1; 0 \} = 0$$

$$LRC_2 = FCF_2 + CSM_2 = 7 + 2,2 = 9,2$$

$$C_2 = C_1 - Y_2^{act} - U_1 = 22 - 8 - 3,7 = 10,3$$

$$U_2 = \bar{Y}_{1,2}^{III} + \Delta CSM_2^* - Y_2^{act} - L_2 = 7 + 2,2 - 8 - 0 = C_2 - LRC_2 = 10,3 - 9,2 = 1,2$$

$$\underline{t = 3}$$

$$FCF_3 = 0$$

$$\Delta FCF_3^{future} = 0$$

$$\Delta FCF_3^{current} = \bar{Y}_{2,3}^{III} - Y_3^{act} = 7 - 7 = 0$$

$$CSM_3^* = \max \{ CSM_2 + \Delta FCF_3^{future}; 0 \} = \max \{ 2,2 - 0; 0 \} = 2,2$$

$$\Delta CSM_3^* = CSM_3^* / (T - t + 1) = 2,2 / (3 - 3 + 1) = 2,2$$

$$CSM_3 = CSM_3^* - \Delta CSM_3^* = 2,2 - 2,2 = 0$$

$$L_3 = -\min \{ CSM_2 + \Delta FCF_3^{future}; 0 \} = -\min \{ 2,2 - 0; 0 \} = 0$$

$$LRC_3 = FCF_3 + CSM_3 = 0$$

$$C_3 = C_2 - Y_3^{act} - U_2 = 10,3 - 7 - 1,2 = 2,2$$

$$U_3 = \bar{Y}_{2,3}^{III} + \Delta CSM_3^* - Y_3^{act} - L_3 = 7 + 2,2 - 7 - 0 = C_3 - LRC_3 = 2,2 - 0 = 2,2$$